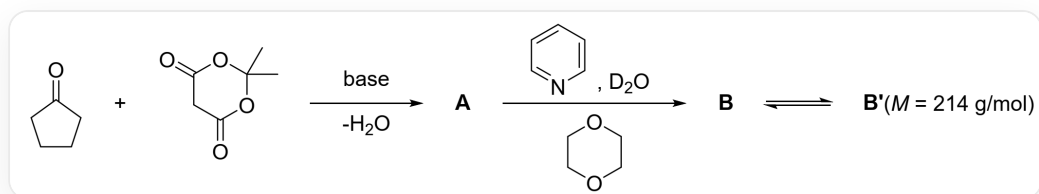


Question



Organic reaction, the substrates are O=C1CCCC1 and O=C(C1)OC(C)(C)OC1=O. The two undergo dehydration under basic conditions to form A. A reacts under the conditions of pyridine, D_2O , and 1,4-dioxane to form B. B and B' constitute an equilibrium, and the molecular weight of B' is 214 g/mol

It is known that B does not contain a hydroxyl group, and B and B' are tautomers. B' generates G and H upon heating, and the reaction mechanism may be as follows: B' loses $(CH_3)_2CO$ under heating conditions to form C. C loses CO_2 to form D. D loses CO to form E. E can be converted to F. F can be converted to G and H.

It is known that the reaction types for B' losing small molecules to form C and F generating G and H are the same. D has two carbon atoms adopting sp hybridization, E has an electron-deficient structure, F has a six-membered ring, and the molecular weight of G is greater than that of H.

Which of the following options is correct?

- A. If isotopic differences are ignored, B has one more H_2O in its molecular formula than A.
- B. B' has a conjugated diene structure and lacks a hydroxyl group.
- C. There are no atoms in C that adopt sp hybridization.
- D. D contains a carbon-carbon triple bond structure
- E. E exhibits a carbene structure and the carbon that does not satisfy the octet rule conjugates with a carbon-carbon double bond.

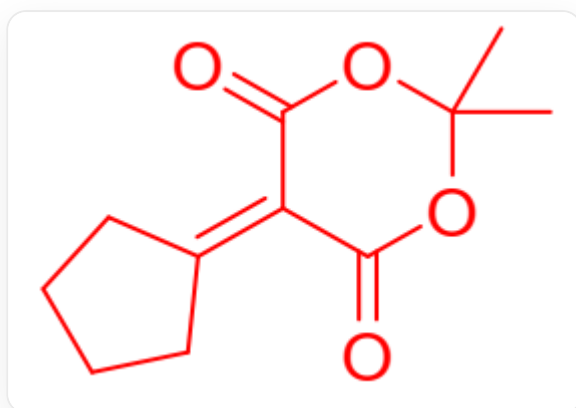
- F.** The structures of D, E, F, G, and H all contain mirror planes (assuming the rings are planar).

Answer

Correct Answer: **F**

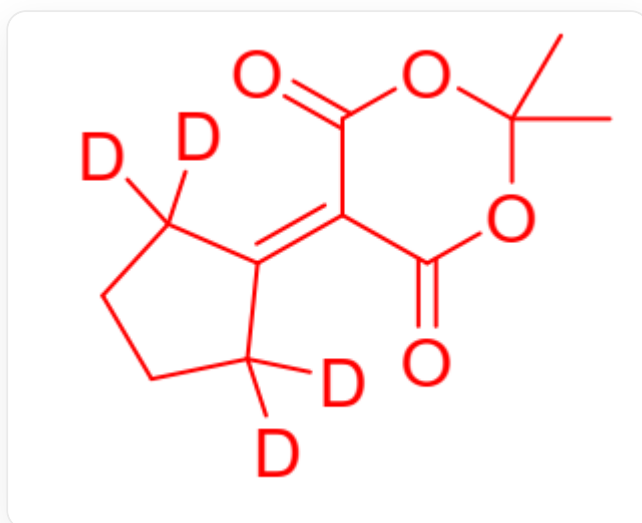
Detailed Explanation

A=



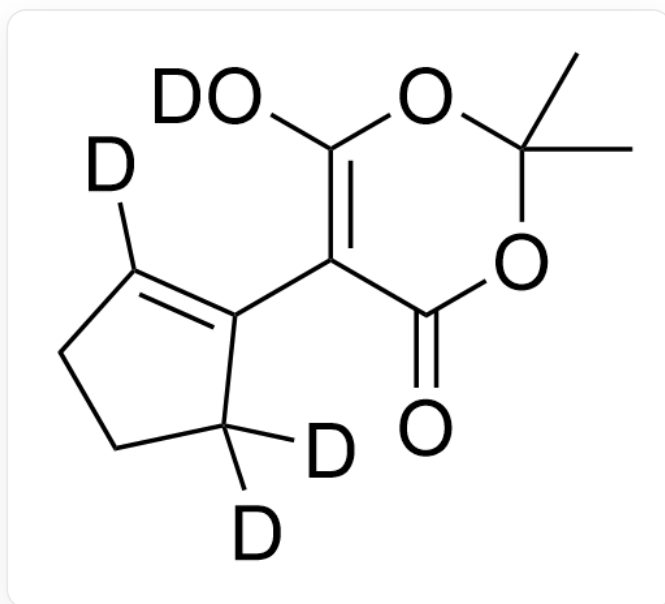
CC1(C)OC(=O)C(=C2CCCC2)C(=O)O1

B=



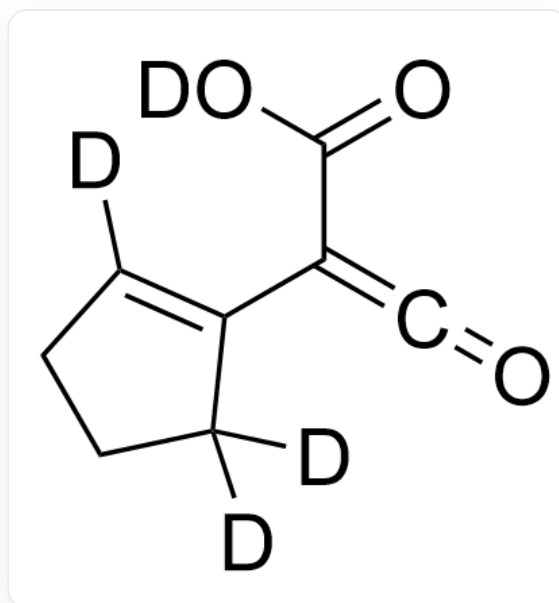
CC1(C)OC(=O)C(=C2C([2H])([2H])CCC2([2H])[2H])C(=O)O1

B'=



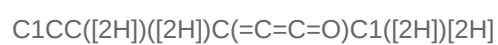
CC1(C)OC([2H])=C(C2=C([2H])CCC2([2H])[2H])C(=O)O1

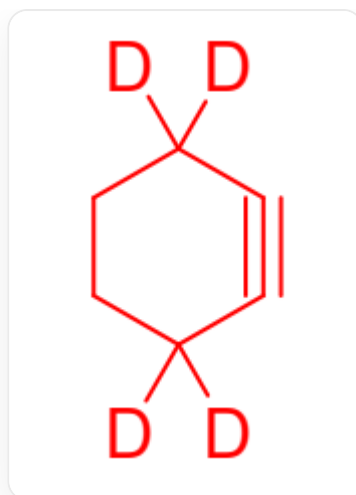
C=



C1CC([2H])([2H])C(=C1[2H])C(=C=O)C(=O)O[2H]

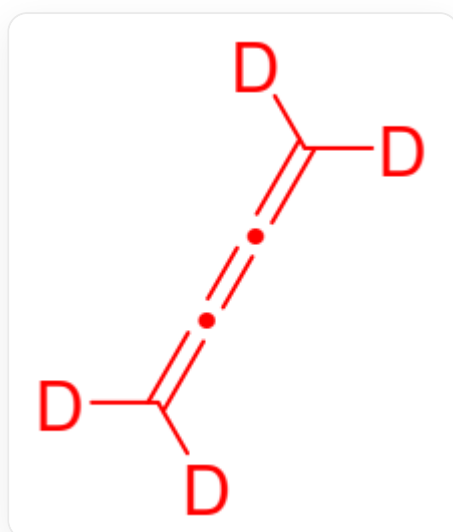
D=

[2H]C1([2H])CCC([2H])([2H])C1=[C]
$$F =$$



[2H]C1([2H])CCC([2H])([2H])C#C1

G=



C([2H])([2H])=C=C=C([2H])[2H]

$H=C_2H_4$

The acidic hydrogen in A is deuterated to form B. B and B' exhibit keto-enol tautomerism. A 1,2-migration occurs from E to F. The decomposition of F and B involves retro-cycloaddition reactions.

If isotopic differences are ignored, A and B have the same molecular formula.

CHECKPOINT

1 PTS

If isotopic differences are ignored, A and B have the same molecular formula.

B' contains a conjugated diene and a hydroxyl group.

CHECKPOINT

1 PTS

B' contains a conjugated diene and a hydroxyl group.

The carbon atom of the ketene in C is sp-hybridized.

CHECKPOINT

1 PTS

The carbon atom of the ketene in C is sp-hybridized.

D contains an allene structure but no carbon-carbon triple bond.

CHECKPOINT

1 PTS

D contains an allene structure but no carbon-carbon triple bond.

E has a carbene structure, but the non-bonding orbital of the carbene and the orbital forming the carbon-carbon double bond are not parallel, so the carbon violating the octet rule cannot conjugate with the carbon-carbon double bond.

CHECKPOINT

1 PTS

In E, the carbon violating the octet rule cannot conjugate with the carbon-carbon double bond.

If the rings of D, E, and F are assumed to be planar, the ring plane serves as a molecular mirror plane; molecules G and H are also planar and possess mirror planes. Option F is correct.

CHECKPOINT

1 PTS

If the rings are assumed planar, the structures of D, E, F, G, and H all contain mirror planes.